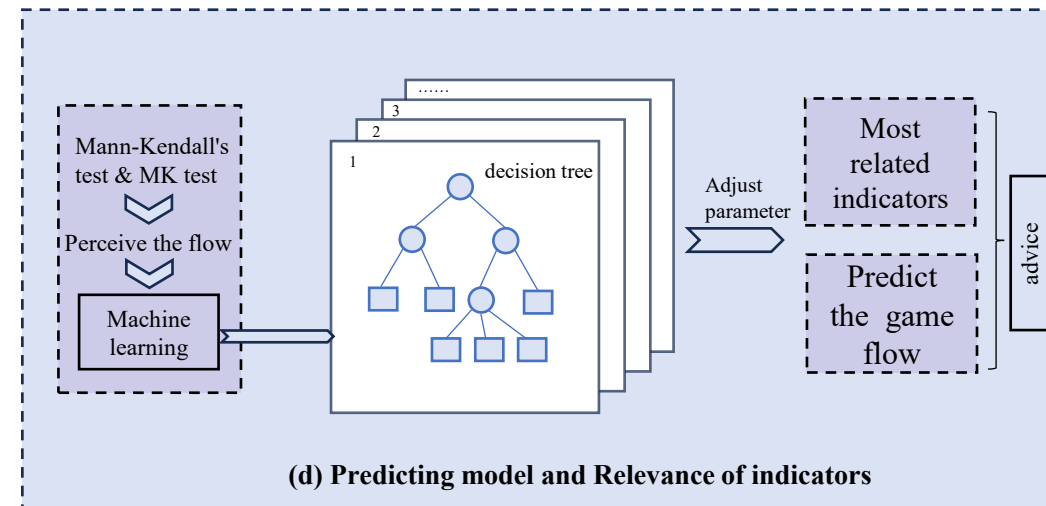
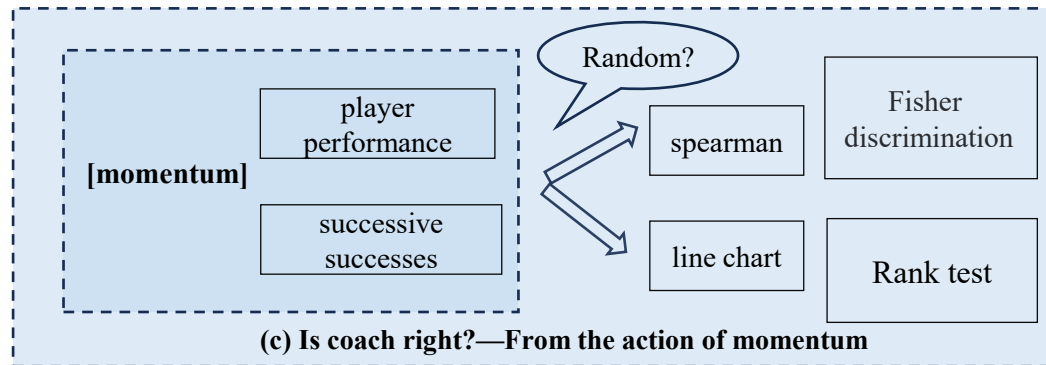
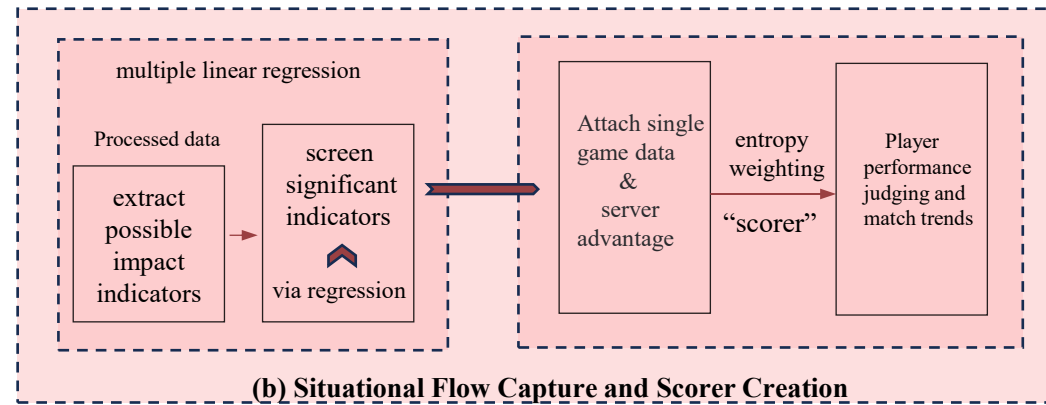
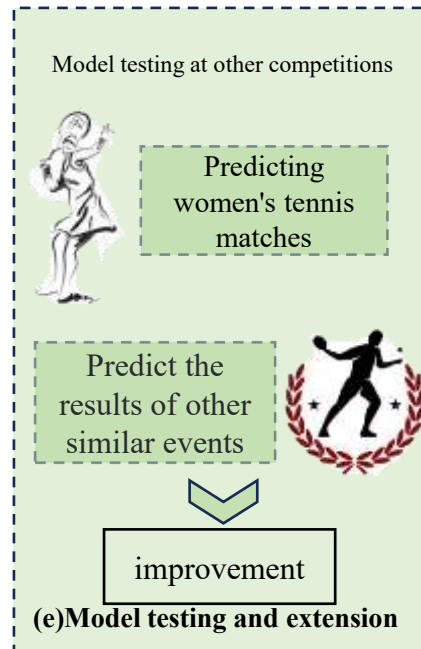
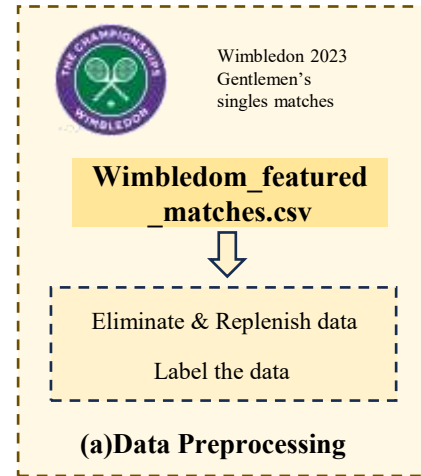




Smartphone Battery Modeling Research Paper Workflow

(a) Problem & Data Input



Smartphone
Battery Problem



AndroWatts
Dataset

(b) Battery Framework

EXTERNAL FACTORS

CPU Display WiFi GPS

INTERNAL FACTORS

Temperature, Capacity Fade, Voltage

(c) Three-Layer Model Framework



Model I
Battery State



Model II
Power Consumption



Model III
Aging



Unified Model

(d) Application & Analysis



TTE
Prediction



Sensitivity
Analysis



Monte Carlo
Simulation

(e) Recommendations & Extensions



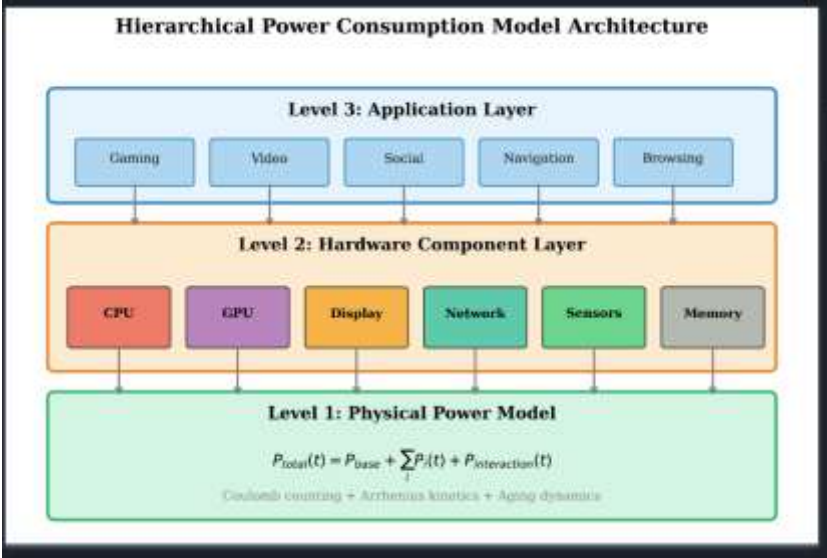
User
Suggestions



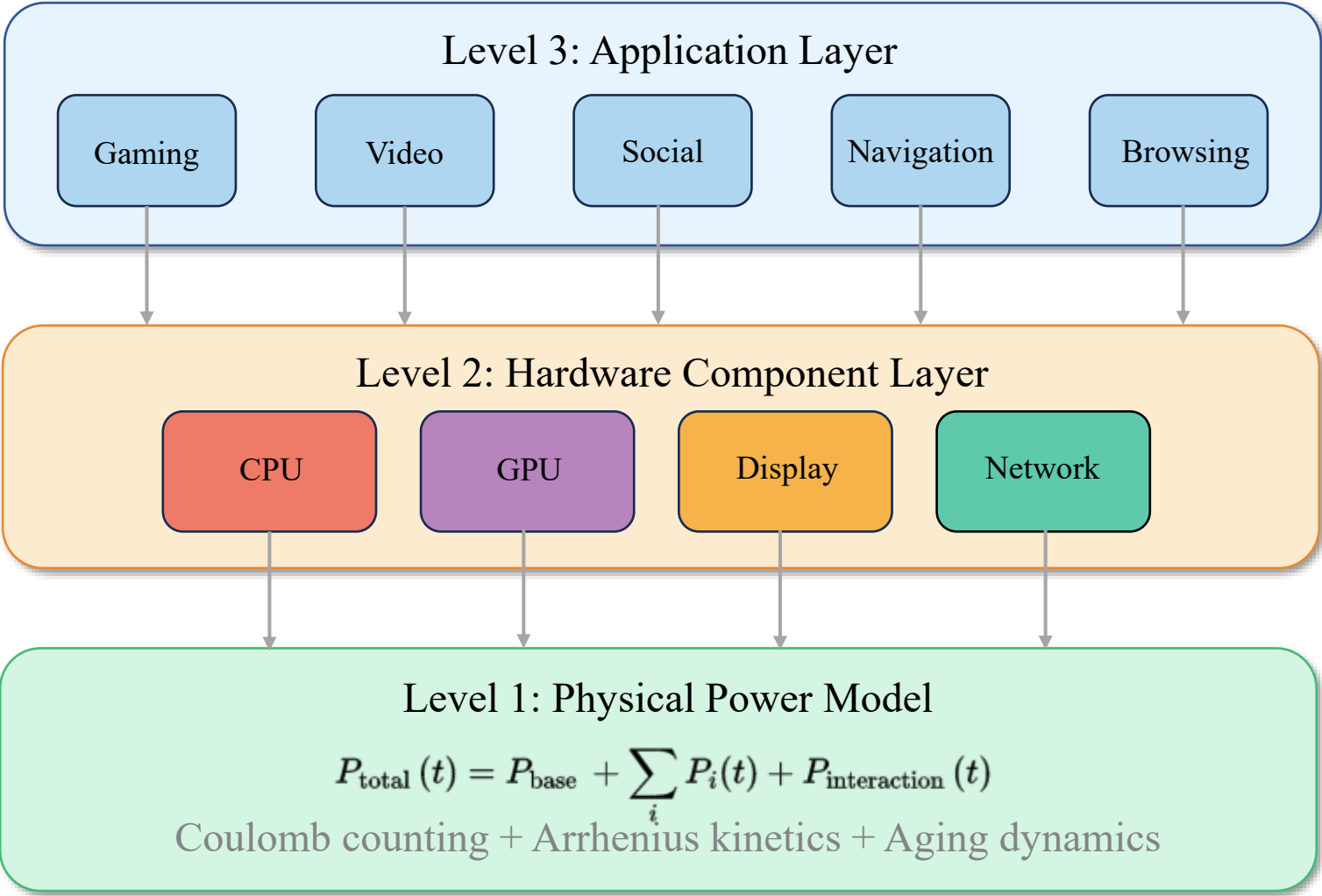
OS
Strategies

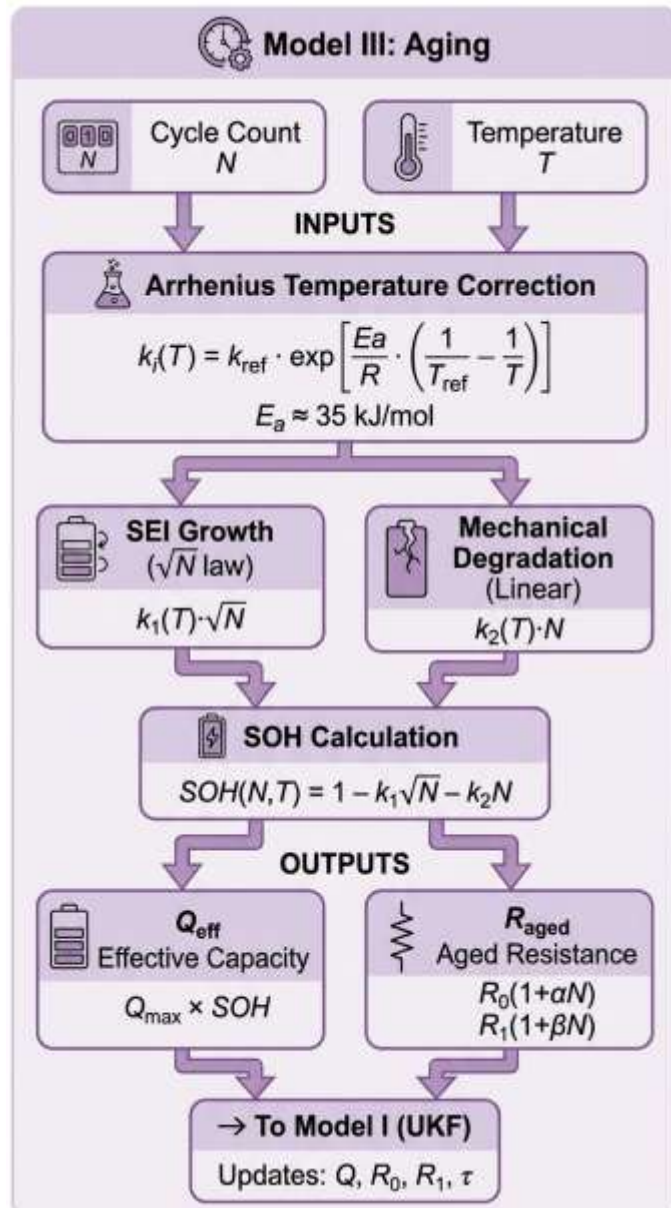


Device
Extension



Hierarchical Power Consumption Model Architecture





Smartphone Battery Modeling Research Paper Workflow

